

Machine Learning (CAI-262)

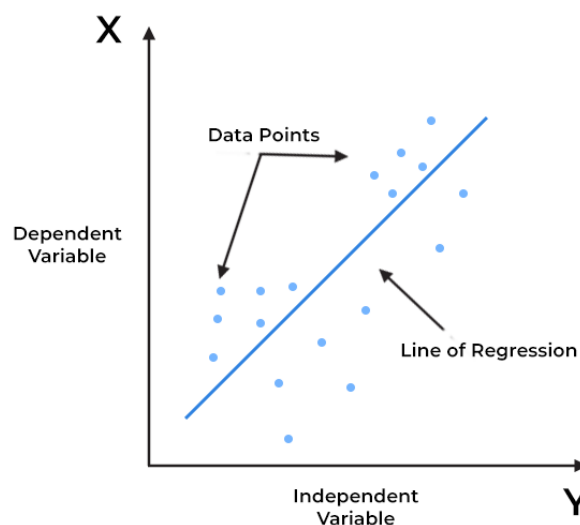
Objectives:

- Linear Regression
 - Types of Regression
 - Simple Linear Regression derivation
-

Linear regression

Linear regression analysis is used to predict the value of a variable based on the value of another variable. The variable you want to predict is called the **dependent** variable. The variable you are using to predict the other variable's value is called the **independent** variable.

This form of analysis estimates the coefficients of the linear equation, involving one or more independent variables that best predict the value of the dependent variable. Linear regression fits a straight line or surface that minimizes the discrepancies between predicted and actual output values. There are simple linear regression calculators that use a “least squares” method to discover the best-fit line for a set of paired data. You then estimate the value of X (dependent variable) from Y (independent variable).



Linear regression is an algorithm that provides a linear relationship between an independent variable and a dependent variable to predict the outcome of future events. It is a statistical method used in data science and machine learning for predictive analysis. Thus, linear regression is a supervised learning algorithm that simulates a mathematical relationship between variables and makes predictions for continuous or numeric variables such as sales, salary, age, product price, etc.

Best Fit Line for a Linear Regression Model

In the above figure,

X-axis = Independent variable

Y-axis = Output / dependent variable

Line of regression = Best fit line for a model

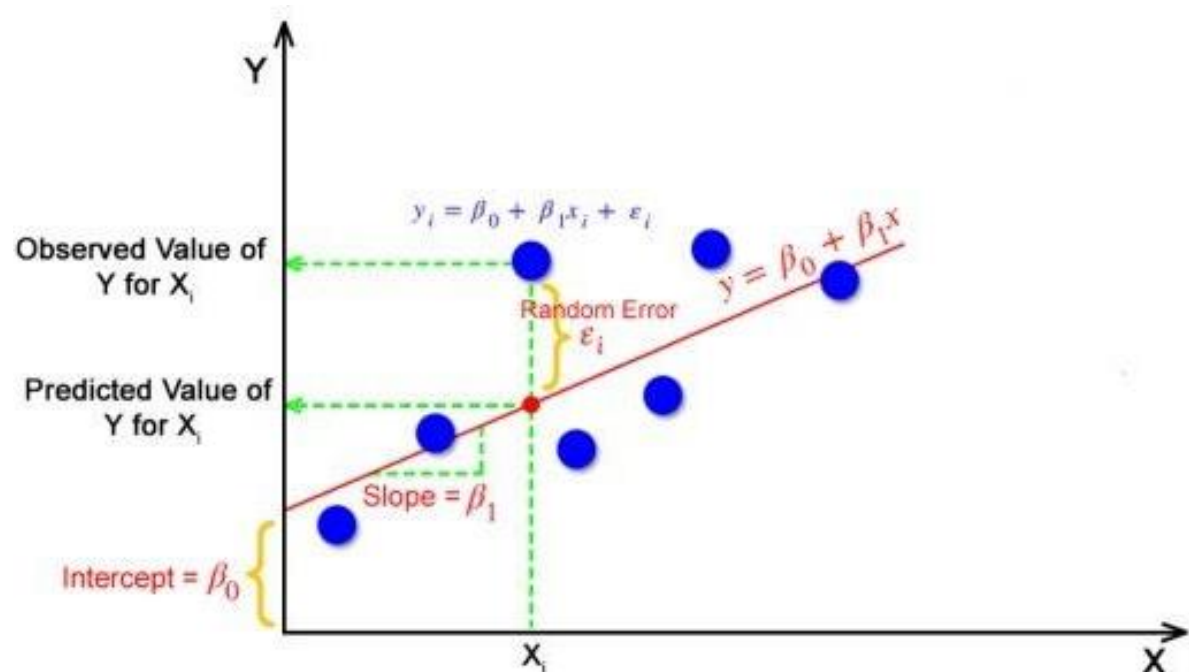
Here, a line is plotted for the given data points that suitably fit all the issues. Hence, it is called the '**best fit line.**' The goal of the linear regression algorithm is to find this best fit line seen in the above figure.

Cost Function of Linear Regression

The cost function helps to work out the optimal values for B_0 and B_1 , which provides the best fit line for the data points. In Linear Regression, generally **Mean Squared Error (MSE)** cost function is used, which is the average of squared error that occurred between the $y_{\text{predicted}}$ and y_i .

We calculate MSE using simple linear equation $y=mx+b$:

$$MSE = \frac{1}{N} \sum_{i=1}^n (y_i - (\beta_1 x_i + \beta_0))^2$$



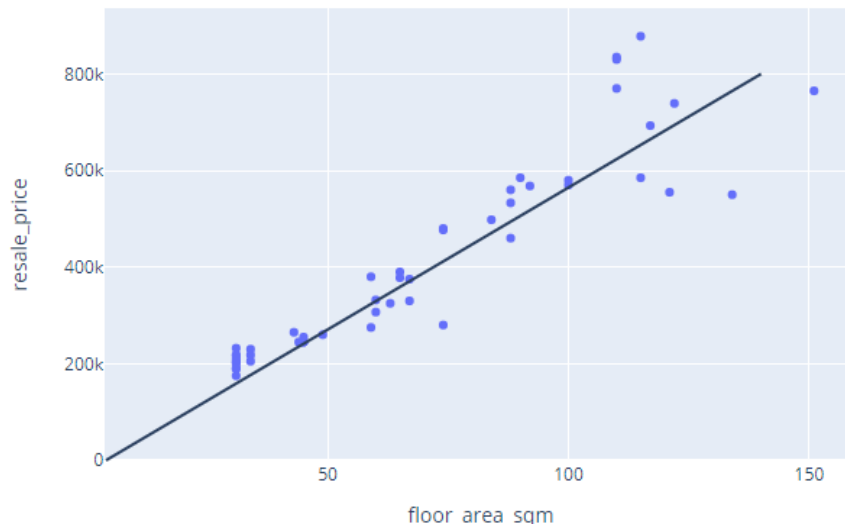
Types of Regression:

There are mainly two different types of linear regression, **Simple Linear Regression** (a.k.a Univariate Linear Regression) and **Multiple Linear Regressions** (a.k.a Multivariate Linear Regression).

As the name suggest, in a simple linear regression, you are trying to find the linear relationship between one independent variable and one dependent variable.

Whereas in a multiple linear regression, you are trying to find the linear relationship between two or more independent variables and one dependent variable.

Example of Simple Linear Regression:



Simple Linear Regression

So as I mentioned above, the goal of simple linear regression model or technique is to find some kind of linear relationship (or to fit a line) between an independent variable and a dependent variable. The formula to plot that linear relationship line in a simple linear regression would be:

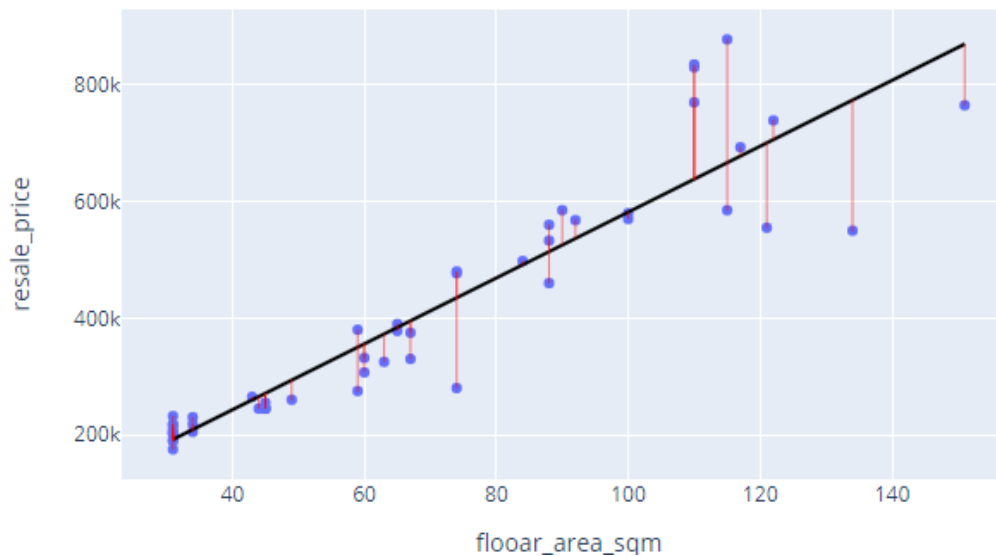
$$y = \beta_0 + \beta_1 x + \varepsilon$$

In the formula:

- **y** is the dependent variable or the output variable you are trying to predict
- **x** would be the independent variable or your input variable
- **β_0** would be your y-intercept, which is the value where the line cross or intersects the y-axis.
- **β_1** would be the slope of the line.
- **ε** would be random error and is a negligible amount.

So now that we have learned about the formula of a simple linear regression, you must be wondering how would you find the value of β_0 and β_1 ! That is, how would you find the slope and the y-intercept to plot the best fit line in your scatter plot (or in your data set), you certainly can't just eyeball it. Infact there are a few different methods to do that and we would use the **Ordinary Least Squares (OLS)** method.

Firstly, what is “**best fit line**”. It simply means to have a line that is able to best reflects the relationship between the dependent and independent variables. If you are using the OLS method, it does so by minimising the squared distances of **residuals**.



If you take a look at the scatter plot above, there are red lines between the data point and the regression line. These red lines are the residual lines, which shows the difference between the real and the predicted values. So in Ordinary Least Square, the aim is to have a line that minimises these residual values.

Proof: Ordinary least squares for simple linear regression

Theorem: Given a [simple linear regression model](#) with independent observations

$$y = \beta_0 + \beta_1 x + \varepsilon, \varepsilon_i \sim \mathcal{N}(0, \sigma^2), i = 1, \dots, n, \quad (1)$$

the parameters minimizing the [residual sum of squares](#) are given by

$$\begin{aligned} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= \frac{s_{xy}}{s_x^2} \end{aligned} \quad (2)$$

where $\bar{x}$ and $\bar{y}$ are the [sample means](#), s_x^2 is the [sample variance](#) of x and s_{xy} is the [sample covariance](#) between x and y .

Proof: The residual sum of squares is defined as

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 . \quad (3)$$

The derivatives of $\text{RSS}(\beta_0, \beta_1)$ with respect to β_0 and β_1 are

$$\begin{aligned} \frac{d\text{RSS}(\beta_0, \beta_1)}{d\beta_0} &= \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i)(-1) \\ &= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) \\ \frac{d\text{RSS}(\beta_0, \beta_1)}{d\beta_1} &= \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i)(-x_i) \\ &= -2 \sum_{i=1}^n (x_i y_i - \beta_0 x_i - \beta_1 x_i^2) \end{aligned} \quad (4)$$

and setting these derivatives to zero

$$\begin{aligned} 0 &= -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \\ 0 &= -2 \sum_{i=1}^n (x_i y_i - \hat{\beta}_0 x_i - \hat{\beta}_1 x_i^2) \end{aligned} \quad (5)$$

yields the following equations:

$$\begin{aligned} \hat{\beta}_1 \sum_{i=1}^n x_i + \hat{\beta}_0 \cdot n &= \sum_{i=1}^n y_i \\ \hat{\beta}_1 \sum_{i=1}^n x_i^2 + \hat{\beta}_0 \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i y_i . \end{aligned} \quad (6)$$

From the first equation, we can derive the estimate for the intercept:

$$\begin{aligned} \hat{\beta}_0 &= \frac{1}{n} \sum_{i=1}^n y_i - \hat{\beta}_1 \cdot \frac{1}{n} \sum_{i=1}^n x_i \\ &= \bar{y} - \hat{\beta}_1 \bar{x} . \end{aligned} \quad (7)$$

From the second equation, we can derive the estimate for the slope:

$$\begin{aligned}
 \hat{\beta}_1 \sum_{i=1}^n x_i^2 + \hat{\beta}_0 \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i y_i \\
 \hat{\beta}_1 \sum_{i=1}^n x_i^2 + (\bar{y} - \hat{\beta}_1 \bar{x}) \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i y_i \quad (7) \\
 \hat{\beta}_1 \left(\sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i \right) &= \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i \\
 \hat{\beta}_1 &= \frac{\sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i} .
 \end{aligned} \tag{8}$$

Note that the numerator can be rewritten as

$$\begin{aligned}
 \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} \\
 &= \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} - n \bar{x} \bar{y} + n \bar{x} \bar{y} \\
 &= \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i - \bar{x} \sum_{i=1}^n y_i + \sum_{i=1}^n \bar{x} \bar{y} \\
 &= \sum_{i=1}^n (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}) \\
 &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})
 \end{aligned} \tag{9}$$

and that the denominator can be rewritten as

$$\begin{aligned}
 \sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i^2 - n \bar{x}^2 \\
 &= \sum_{i=1}^n x_i^2 - 2n \bar{x} \bar{x} + n \bar{x}^2 \\
 &= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2 \\
 &= \sum_{i=1}^n (x_i^2 - 2\bar{x} x_i + \bar{x}^2) \\
 &= \sum_{i=1}^n (x_i - \bar{x})^2 .
 \end{aligned} \tag{10}$$

With (9) and (10), the estimate from (8) can be simplified as follows:

$$\begin{aligned}
 \hat{\beta}_1 &= \frac{\sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i} \\
 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\
 &= \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \\
 &= \frac{s_{xy}}{s_x^2} .
 \end{aligned} \tag{11}$$

Example:

Formula for linear regression equation is given by:

$$y = a + bx$$

a and b are given by the following formulas:

$$a (\text{intercept}) = \frac{\sum y \sum x^2 - \sum x \sum xy}{(\sum x^2) - (\sum x)^2}$$

$$b (\text{slope}) = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

Where,

x and y are two variables on the regression line.

b = Slope of the line.

a = y -intercept of the line.

x = Values of the first data set.

y = Values of the second data set.

Question: Find linear regression equation for the following two sets of data:

x	2	4	6	8
y	3	7	5	10

Solution:

Construct the following table:

x	y	x^2	xy
2	3	4	6
4	7	16	28
6	5	36	30
8	10	64	80
$\sum x$ = 20	$\sum y$ = 25	$\sum x^2$ = 120	$\sum xy$ = 144

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$b = \frac{4 \times 144 - 20 \times 25}{4 \times 120 - 400}$$

$$b = 0.95$$

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$a = \frac{25 \times 120 - 20 \times 144}{4(120) - 400}$$

$$a = 1.5$$

Linear regression is given by:

$$y = a + bx$$

$$y = 1.5 + 0.95x$$